Prior-Art Validation Matrix

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Prior-Art Validation Matrix

This is a technical landscape review, not a legal opinion or a substitute for a professional patentability/FTO search. Google Patents status labels are not legal conclusions. Dates and family status must be confirmed in official registers.

1. Executive validation result

The original broad proposition, “blast-furnace digital twin combining physics and ML to diagnose root cause and recommend corrective action,” is **not novel enough to claim broadly**.

The strongest references are:

- **EP4154105A1 / EP4154105B1 (Tata Consultancy Services)**: self-organizing cyber-physical industrial plant monitoring with hybrid physics/data models, time-lag identification, regime analysis, real-time root-cause analysis, hidden-variable soft sensors, and corrective actions, including steel and blast-furnace examples.
- **CN122654903A (Central South University)**: blast-furnace root-cause analysis combining expert physical knowledge, causal structure, hierarchical lag estimation, Granger causality, and confidence-scored causal paths.
- **US11899527B2 (Caterpillar)**: reduced-order digital-twin anomaly detection and root-cause identification by comparing measured data with fault models.
- **US12321163B2 / US20230057291A1 (Daicel)**: causal production-facility prediction using historical process data and residence-time alignment for explanatory and response variables.
- **CN109918702A (Shanghai Jiao Tong University)**: blast-furnace hybrid neural-network/mechanistic mass/heat constrained multi-objective optimization.

Accordingly, a defensible filing should emphasize the **specific combination and sequence**, not its generic components:

1. required-production-conditioned empirical/physical Golden Bands;
2. physical lag-active comparison against both those bands and an excluded-recent-window rolling reference;
3. detection specifically during an unchanged-manipulated-input interval;
4. fixed baseline sensor-anchor bias plus exact conservation identities and zero-change identity;
5. derivative-preserving soft saturation;
6. candidate gating by both a joint-density envelope/local trust region and multiple named furnace production ceilings; and
7. automatic conversion of a blocked candidate into an MoC trial dossier and separate execution-confirmation ledger.

Even this combination faces a **material obviousness/inventive-step risk** because an examiner can combine TCS, Central South University, Daicel, AspenTech, and known Mahalanobis techniques.

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2. Closest patent and non-patent references

Ref.	Publication / source	Priority / publi-cation	Applicant	Relevant disclosure	Risk
R1	EP4154105A1, granted as EP4154105B1	priority 19 May 2020; A1 29 Mar 2023; B1 2 Jul 2025	Tata Consultancy Services Ltd	Industrial cyber-physical/digital-twin platform; physics, ML, and hybrid models; time-lag identifica-tion; regime shifts; hidden variables; real-time root-cause analysis; operator guidance and corrective ac-tions; steel/blast-furnace use cases.	**Very high**
R2	CN122654903A	filed 4 Jun 2026; pub-lished 28 Aug 2026	Central South University	Blast-furnace ab-normal-condition root cause using expert physical knowledge, Granger causality, adaptive group lasso, hierarchical lag search, physi-cal lag priors, path search, and confi-dence scoring.	**Very high**
R3	US11899527B2 / US20230168961A1	priority 30 Nov 2021; A1 1 Jun 2023; B2 13 Feb 2024	Caterpillar Inc.	Sensor-fed re-duced-order digi-tal twin; anomaly from model/measured residual; multiple fault models; se-lect closest fault output as root cause.	High
R4	US12321163B2 / US20230057291A1	filed 4 Feb	Daicel Corporation	Causality informa-tion in a produc-	High

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Ref.	Publication / source	Priority / publication	Applicant	Relevant disclosure	Risk
		2021; A1 23 Feb 2023; B2 3 Jun 2025		tion-facility prediction model; positive/negative direction; timing alignment using residence time; optimum manufacturing conditions.	
R5	CN111831719A	priority 22 Jul 2020; published 27 Oct 2020	Shandong Iron and Steel Co.	Blast-furnace online data, operation/raw-material division, digital/mechanistic model corrected by association analysis, deep learning and genetic-algorithm control.	High
R6	CN109918702A	priority 3 Jan 2019; published 21 Jun 2019	Shanghai Jiao Tong University	Blast-furnace burden/operation hybrid neural + mass/heat constrained multi-objective optimization for fuel, emissions, cost, and metal quality.	High
R7	WO2021076760A1	priority 18 Oct 2019; published 22 Apr 2021	Aspen Technology Inc.	Automated model development from historical plant tags; informative moves; multivariable predictive control; model growth/calibration; steady-state optimization.	Medium-high
R8	EP4567667A1	priority 5 Dec	British Telecommunications	Modular digital twin with learning	High for generic hy-

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Ref.	Publication / source	Priority / publication	Applicant	Relevant disclosure	Risk
		2023; published 11 Jun 2025	PLC	and differentiable physics blocks, residual block, sensor data, explanations, and control output.	brid twin
R9	WO2024141260A1	priority 27 Dec 2022; published 4 Jul 2024	Siemens Schweiz AG	Real-time sensor-based digital-twin calibration using error gradient and candidate parameter optimization.	Medium-high
R10	US20170286572A1	priority 31 Mar 2016; published 5 Oct 2017	General Electric Co.	General digital twin of a twinned physical system using sensor/operating data.	Medium
R11	US20240337486A1	publication 10 Oct 2024	—	Learned rules for sensor-reading correction and missing-data enhancement.	Medium
R12	US6486661B2	priority 14 Feb 2001; expired	—	Generic sensor bias-drift compensation.	Low-medium
R13	US9799110B2 / US20160148365A1	JP priority 29 Jul 2013; A1 26 May 2016; B2 24 Oct 2017	JFE Steel Corp.	Blast-furnace abnormality detection from multiple tuyere images using PCA and distance from normal subspace.	Medium

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Ref.	Publication / source	Priority / publication	Applicant	Relevant disclosure	Risk
R14	CN1403593A, granted CN1224720C	priority 17 Oct 2002; published 19 Mar 2003	Zhejiang University	Online blast-furnace data, expert intelligent control, fault charts, foreman guidance, best thermal-state operation.	High for broad BF expert-system claims
R15	CN1038146C	older Chinese BF expert-system family	—	Sensor collection, fuzzy expert knowledge, furnace-state and abnormal-condition prediction/control.	High for broad BF diagnosis
R16	US20060230018A1	priority 8 Apr 2005; published 12 Oct 2006; abandoned	Caterpillar Inc.	Mahalanobis distance and genetic algorithm for variable selection and separation of normal/abnormal data.	Medium for generic Mahalanobis claims
R17	US20180283815A1 / US10663238B2	priority 28 Mar 2017; A1 4 Oct 2018	Honeywell UOP LLC	Historical/current plant data, operating boundaries, relationships among operating parameters, target optimization, maldistribution detection.	Medium-high
R18	US10935962B2 / US20170153630A1	priority 30 Nov 2015	National Cheng Kung University	Two-phase production yield-loss root-cause ranking by process tool/stage and parameters.	Medium
R19	Caers and Eskanlou, reverse-flotation causal study cited in	2026	Academic paper	Demonstrates predictive/causal	High for generic

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Ref.	Publication / source	Priority / publi-cation	Applicant	Relevant disclosure	Risk
	`knowledge/09-confounding-and-causality.md`			disagreement, backdoor, g-com-putation, DML, lag, and non-identifiability.	causal-audit concepts
R20	Blast-furnace causality by conver-gent cross mapping, DOI 10.1080/03019233.2022.2065447	2022	Academic paper	Causal relation-ship analysis and blast-furnace state prediction.	Medium-high

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3. Element-by-element challenge of proposed independent claim

Proposed element	Closest teaching	Anticipated examiner position	Proposed distinction
Read-only sensor/historian acquisition	R1, R5, R7, R14	Routine plant data acquisition; read-only choice may be obvious security design.	Retain as safety architecture, but do not rely on it alone for novelty.
Mutually exclusive causal-role assignment	R4 causal explanatory/response variables; R1 model types and plant variables	Variable classification is known.	Require five roles and use the roles to control attribution, absolute-value withholding, and decision-model feature eligibility.
Production-target-conditioned empirical bands	R17 operating boundaries; R4 setting/management ranges; R7 steady-state targets	Historical target ranges are known.	Require permeability-qualified blast-furnace cohorts plus physics sensitivity shifting for requested hot-metal tonnage across all roles. Still medium risk.
Per-tag physical lag convolution	R1 time-lag identification; R2 hierarchical lag with physical priors; R4 residence-time alignment	Lags in process plants and blast furnaces are known.	Require heterogeneous gas/injectant/burden/inventory kernels and dual comparison against Golden Band plus excluded-recent rolling reference during an unchanged-input interval.
Detect shortfall while manipulated inputs stay in band	R1 performance dip/root cause; R2 abnormal furnace root cause	Broad performance-dip diagnosis is known.	Central narrowing feature. Explicitly require unchanged manipulated inputs and rank disturbance/state-response deviations.
Sensor-anchor bias held fixed over counterfactual	R3 residual anomaly; R9 real-time calibration; R11 sensor correction	Sensor/model reconciliation is known.	Require fixed baseline additive anchor, exact pressure-conservation identity, and zero-change output identity. Do not claim generic calibration.
Smooth saturation preserving nonzero derivative	Generic numerical methods; R8 differentiable physics	Soft clipping and differentiability are known mathematically.	Claim only in combination with fixed anchor and conservation-consistent blast-furnace counterfactual. Avoid standalone claim.
Predictive vs causal decision residual	R4 causal model; R19 causal-estimation literature	Causal feature restriction is known.	Treat as dependent limitation; not likely a standalone inventive concept.

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Proposed element	Closest teaching	Anticipated examiner position	Proposed distinction
Mahalanobis envelope and local trust region	R16; widespread multivariate process control	Mahalanobis anomaly envelopes are old.	Require simultaneous joint-density envelope, local step distance, and five physical furnace ceilings, leading to automatic MoC dossier rather than controller output.
Five-ceiling binding arbiter	BF physics and optimization prior art such as R6	Individual physical constraints are known.	Claim explicit lowest-ceiling arbitration combined with density/trust and unchanged-sheet attribution.
MoC dossier and audit/execution split	Governance/audit techniques	Likely administrative/business-method objection.	Keep as dependent claim. Tie to technical suppression and model/envelope evidence.

4. Validation of four proposed invention families

FAMILY A: LAG-CONVOLVED TARGET-BAND ATTRIBUTION

Status: AMBER / potentially patentable only when narrowly combined.

R1 and R2 are severe references because they teach industrial/blast-furnace root-cause analysis, time lag, physical knowledge, regime analysis, and corrective guidance. R4 teaches residence-time alignment and causal direction in a production facility. The likely defensible distinction is not “lag-aware root cause” alone, but the sequence of:

- target-production-conditioned permeability-qualified cohort bands;
- unchanged manipulated-input interval;
- per-tag physical convolution;
- dual Golden-Band and prior rolling-reference comparison; and
- production-loss attribution by causal role.

FAMILY B: SENSOR-ANCHORED DERIVATIVE-PRESERVING HYBRID TWIN

Status: AMBER-GREEN / strongest technical nucleus if experimental effects are documented.

R3, R8, R9, R11, and R12 cover digital-twin residuals, physics/learning blocks, calibration, and sensor correction. None identified in this search expressly teaches the fixed baseline anchor held invariant across a counterfactual together with zero-change identity, exact pressure-split conservation, and smooth nonzero-derivative saturation in a blast furnace.

An application should include comparative experiments:

- hard clip versus soft saturation response near a limit;
- free-running versus anchored pressure-drop error;

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- zero-change drift before and after the method; and
- conservation residual before and after reconciliation.

FAMILY C: ENVELOPE/CEILING/MOC GATE

Status: AMBER / obviousness risk.

Mahalanobis envelopes, MPC constraints, blast-furnace hybrid optimization, and physical constraints are known separately. The narrow combination of two Mahalanobis tests, five named BF ceilings, causal-identifiability status, and automatic conversion from routine advisory to a technical trial dossier may survive as a dependent or second-family claim, but a generic optimization-envelope claim is weak.

FAMILY D: EVIDENCE/CAPABILITY CLASSIFIER

Status: RED-AMBER / likely abstract, obvious, or presentation-of-information risk alone.

Claim only as a technical dependent limitation that physically suppresses computation/output of unsupported absolute latent states while retaining differential response. Do not lead with labels or user-interface presentation.

5. Public-disclosure risk

The source repository and interactive demonstration were public on **10 November 2026**.

- The USPTO states an inventor-originated public disclosure can fall within the one-year AIA grace period. The factual authorship/derivation of the disclosure must be documented.
- The USPTO also warns such pre-filing disclosure may preclude foreign patenting.
- India generally applies absolute novelty, subject to narrow statutory exceptions that must be analyzed by Indian counsel. Public GitHub availability is a serious India/PCT risk.
- If any inventor is resident in India, Section 39 and Form 25 must be considered before an outside-India filing.

6. Search limitations

This review used keyword and classification-oriented searches of Google Patents, public patent aggregators, official USPTO/IP India materials, and project literature. It is not:

- a professional classification/subclassification search;
- a prosecution-history or claim-construction opinion;
- a freedom-to-operate opinion;
- a legal-status opinion;
- a search of unpublished applications; or
- a substitute for searches in USPTO Patent Public Search, Espacenet, WIPO Patentscope, Indian Patent Advanced Search System, CNIPA, KIPRIS, J-PlatPat, and commercial semantic databases.

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